

Residual Learning for Agile Locomotion: Application to a Two-Wheeled Legged Robot



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Thesis recap

1 What is Residual Learning

MPC provides an interpretable, physically grounded baseline, but model mismatch, contact uncertainty and real-time computation can limit performance. **RL learns adaptable behavior through experience**, including interactions that are difficult to model. End-to-end policies require careful reward design and substantial training, and offer less interpretability. This thesis combines MPC with a neural network that learns residual corrections for improved locomotion.

Residual Learning Idea

- MPC provides a structured, interpretable, constraint-aware baseline.
- RL adapts to model errors, contacts and terrain variations.
- The policy does not learn locomotion from scratch.
- It learns a correction to the MPC action.
- This combines model-based reliability with learned robustness.



Thesis recap

1 What is Residual Learning

Residual Learning has been widely used to augment model-based locomotion controllers.

In previous works, residual policies are used to:

- compensate for model mismatch.
- handle effects not explicitly modeled in MPC.
- enhance specific modules of the control pipeline.
- improve robustness on complex terrains.
- ...

Most applications focus on humanoids and quadruped robots. In contrast, residual learning for bipedal wheeled-legged robots remains less explored.

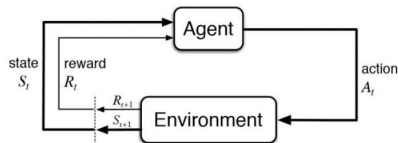
This work applies residual learning to a bipedal wheeled-legged robot, aiming to combine MPC structure with RL adaptability.



Definitions

1 What is Residual Learning

- **Agent:** policy π_ϕ , implemented by a neural network.
- **Observation space:** possible policy inputs, $o_t \in \mathcal{O}$.
- **Action space:** possible outputs, $a_t \in \mathcal{A}$.
- **Reward:** scalar signal r_t evaluating behavior.
- **Transition:** $s_{t+1} \sim P(\cdot \mid s_t, a_t)$, with reward r_t and observation o_{t+1} .
- **Replay buffer:** stored transitions. PPO always empty this buffer and the end of the episode.



- **Simulation frame:** one simulation step.
- **Actor:** policy π_ϕ .
- **Critic:** estimates $V_\theta(s)$ or $Q_\theta(s, a)$.



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Residual MPC: Blending RL with GPU-Parallelized MPC

2 Recap on papers

MLP: three hidden layers, 256 ELU units each; Gaussian policy $\mathcal{N}(\mu, \sigma^2)$.

Scaling λ limits deviations from the nominal controller.

$$o = [p^\top, \theta^\top, q_j^\top, \omega^\top, v^\top, \dot{q}_j^\top, \phi^\top, V_{\text{MPC}}]^\top \in \mathbb{R}^{54}.$$

Joint action – joint blending

$$\tau = K_p(q_{\text{MPC}} + \lambda a - q) + K_d(\dot{q}_{\text{MPC}} - \dot{q}) + \tau_{\text{MPC}}.$$

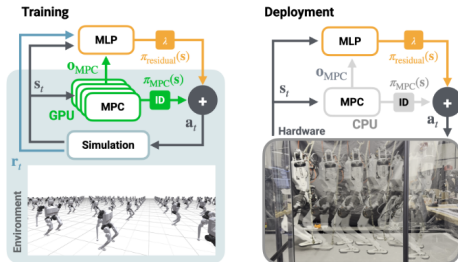
Joint action – torque blending

$$\tau = \tau_{\text{MPC}} + \lambda \tau_{\text{res}},$$

$$\tau_{\text{res}} = K_p(\hat{q} + \lambda a - q) - K_d \dot{q}.$$

Torque action – torque blending

$$\tau = \tau_{\text{MPC}} + \lambda a.$$



p : base position; θ : orientation; q_j : joint positions; ω, v : angular/linear velocities; ϕ : contact phase; V_{MPC} : MPC value estimate.

Author suggestion: Joint-torque blending achieves similar performance to joint-joint blending while avoiding corrections relative to potentially infeasible q_{MPC} setpoints when MPC diverges.

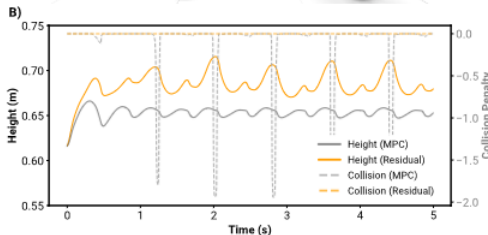
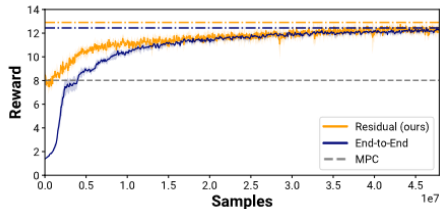
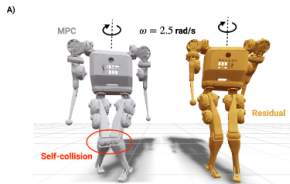


Residual MPC: training and self-collision avoidance

2 Recap on papers

Reward obtained during training with the 3 control techniques: MPC-only, Residual Learning, E2E approach

Given and angular command to the humanoid robot, with MPC approach, there is a knee-collision which is solved with Residual Learning approach





Residual MPC: terrain adaptation

2 Recap on papers

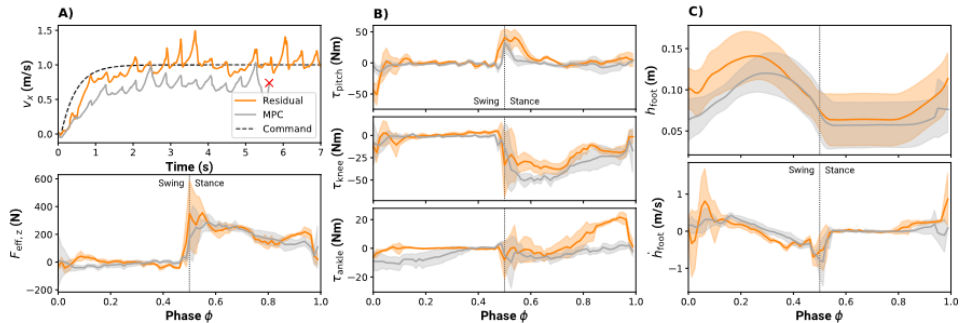


Fig. 17. Analysis of residual and MPC data from locomotion over uneven terrain at 1.0 m/s. **A)** Velocity tracking and effective vertical commanded force for the right foot. **B)** Sagittal joint torques through the phase. The residual policy exhibits emergent toe-off and heel touchdown behavior from training. **C)** Height and velocity profiles of the center of the foot link during swing and stance.



Residual MPC: robustness and achievable velocities

2 Recap on papers

- The residual policy avoids unmodeled self-collisions.
- It handles unseen uneven, rough and sloped terrain without fine-tuning or domain randomization.
- Achievable velocities improve by 78% in v_x , 12% in v_y , and 9% in ω_z .

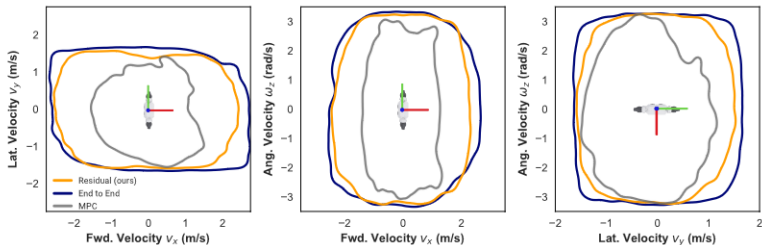
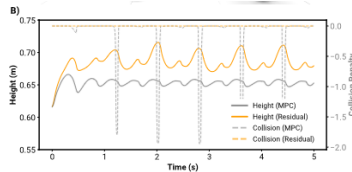


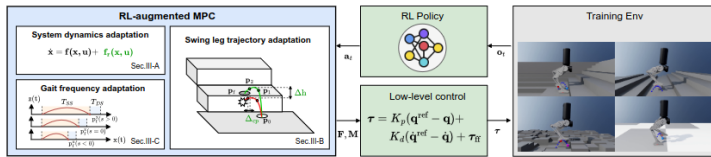
Fig. 10. 95% kernel density estimate boundaries of the trackable velocities for each policy. The residual policy network substantially improves the viable range of commands over the MPC baseline.





RL-augmented Adaptive MPC for Bipedal Locomotion

2 Recap on papers



g_p : projected gravity; v_b, ω_b : base velocities; v_{cmd} : velocity command; $q_j, \dot{q}_j$: joint state; a_{t-1} : previous action; ϕ : swing phase; p_{fh} : planned foothold; p_f : foot position; $p_{f,ref}$: reference foot position.

Fig. 2: System architecture of our proposed method. The RL policy augments three key modules in the single rigid body MPC. The low level controller then tracks the contact force $\mathbf{F}$ and moment $\mathbf{M}$ computed by MPC to generate joint torque commands. We leverage the whole-body dynamics of the robot to train the policy across diverse terrains, as illustrated on the right side of the figure.

- MLP: (512, 256, 128) hidden units with ELU activations.
- Observation:

$$o = [g_p^\top, v_b^\top, \omega_b^\top, v_{cmd}^\top, q_j^\top, \dot{q}_j^\top, a_{t-1}^\top, \phi^\top, p_{fh}^\top, p_f^\top, p_{f,ref}^\top]^\top \in \mathbb{R}^{59}.$$

Outputs: residual dynamics $f \in \mathbb{R}^{12}$; swing-foot height Δh ; swing-shape correction Δc_p ; sampling coefficient s controlling gait duration.



Adaptive MPC: success rate and tracking

2 Recap on papers

TABLE V: Comparison between the baseline MPC and our approach. The table shows success rate (SR), forward velocity tracking error (e^v), and roll/pitch errors ($e^{\theta_x}, e^{\theta_y}$) collected from 100 episodes on rough terrain and slippery terrain with varying levels of difficulty. The highest success rate and the lowest tracking errors are highlighted in bold.

Env	height (cm) / μ (-)	SR _{MPC} $\uparrow$ (%)	SR _{ours} $\uparrow$ (%)	e^v_{MPC} $\downarrow$ (m/s)	e^v_{ours} $\downarrow$ (m/s)	$e^{\theta_x}_{MPC}$ $\downarrow$ (deg)	$e^{\theta_x}_{ours}$ $\downarrow$ (deg)	$e^{\theta_y}_{MPC}$ $\downarrow$ (deg)	$e^{\theta_y}_{ours}$ $\downarrow$ (deg)
pyramid	8	100 $\pm$ 0	100 $\pm$ 0	0.12 $\pm$ 0.09	0.12 $\pm$ 0.09	0.55 $\pm$ 0.38	0.55 $\pm$ 0.40	1.71 $\pm$ 0.88	1.6 $\pm$ 0.94
stairs	9	48 $\pm$ 5	96 $\pm$ 1.96	0.15 $\pm$ 0.12	0.14 $\pm$ 0.1	0.68 $\pm$ 1.03	0.59 $\pm$ 0.46	1.96 $\pm$ 1.58	1.68 $\pm$ 1.1
(height)	10	1 $\pm$ 0.99	86 $\pm$ 3.47	0.22 $\pm$ 0.17	0.16 $\pm$ 0.13	0.97 $\pm$ 1.87	0.63 $\pm$ 0.58	2.6 $\pm$ 3.03	1.7 $\pm$ 1.16
random	8	99 $\pm$ 0.99	99 $\pm$ 0.99	0.1 $\pm$ 0.09	0.1 $\pm$ 0.09	0.57 $\pm$ 0.39	0.56 $\pm$ 0.4	1.39 $\pm$ 0.84	1.32 $\pm$ 0.88
stairs	9	69 $\pm$ 4.62	98 $\pm$ 1.4	0.12 $\pm$ 0.11	0.12 $\pm$ 0.1	0.66 $\pm$ 0.89	0.58 $\pm$ 0.43	1.59 $\pm$ 1.51	1.37 $\pm$ 0.95
(height)	10	15 $\pm$ 3.57	85 $\pm$ 3.57	0.19 $\pm$ 0.15	0.12 $\pm$ 0.11	0.96 $\pm$ 1.94	0.61 $\pm$ 0.48	2.08 $\pm$ 2.58	1.47 $\pm$ 1.18
stepping	$\mathcal{U}(-5, 5)$	100 $\pm$ 0	100 $\pm$ 0	0.09 $\pm$ 0.08	0.09 $\pm$ 0.08	0.56 $\pm$ 0.39	0.54 $\pm$ 0.34	1.24 $\pm$ 0.64	1.18 $\pm$ 0.6
stones	$\mathcal{U}(-6, 6)$	86 $\pm$ 3.47	96 $\pm$ 1.96	0.1 $\pm$ 0.09	0.1 $\pm$ 0.09	0.6 $\pm$ 0.59	0.57 $\pm$ 0.49	1.35 $\pm$ 1.14	1.25 $\pm$ 0.87
(height)	$\mathcal{U}(-7, 7)$	68 $\pm$ 4.66	89 $\pm$ 3.13	0.12 $\pm$ 0.11	0.11 $\pm$ 0.09	0.67 $\pm$ 1.0	0.61 $\pm$ 0.68	1.58 $\pm$ 1.82	1.28 $\pm$ 0.95
slippery	0.2	100 $\pm$ 0	100 $\pm$ 0	0.09 $\pm$ 0.08	0.09 $\pm$ 0.07	0.59 $\pm$ 0.29	0.73 $\pm$ 0.34	1.11 $\pm$ 0.43	1.4 $\pm$ 0.38
surface	0.15	81 $\pm$ 3.92	100 $\pm$ 0	0.1 $\pm$ 0.08	0.1 $\pm$ 0.07	0.79 $\pm$ 0.94	0.65 $\pm$ 0.34	1.26 $\pm$ 0.76	1.41 $\pm$ 0.4
(μ)	0.1	19 $\pm$ 3.92	100 $\pm$ 0	0.16 $\pm$ 0.12	0.11 $\pm$ 0.07	1.56 $\pm$ 2.97	0.61 $\pm$ 0.37	1.55 $\pm$ 1.7	1.42 $\pm$ 0.42
	0.05	5 $\pm$ 2.18	74 $\pm$ 4.39	0.22 $\pm$ 0.16	0.13 $\pm$ 0.09	1.95 $\pm$ 3.6	0.78 $\pm$ 1.29	1.56 $\pm$ 1.84	1.48 $\pm$ 0.71

Success rate (SR) measures the fraction of successful, non-terminated trials.



Adaptive MPC: challenging terrain

2 Recap on papers

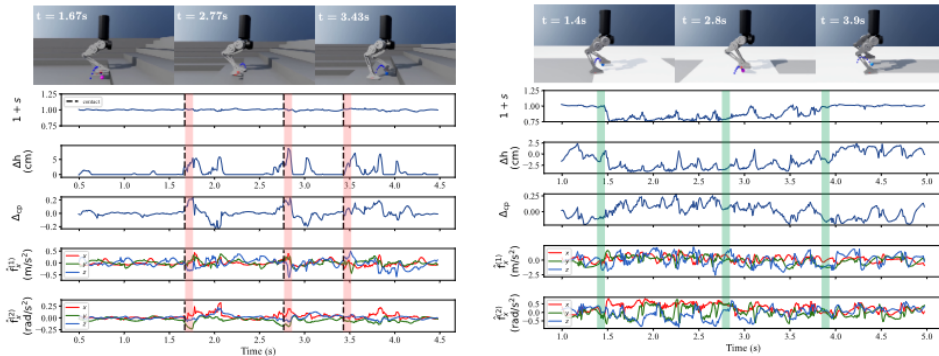


Fig. 3: The figure illustrates actions taken by the RL policy and time-lapse images of the robot at times marked by red (rough terrain instances) and green (slippery terrain instances) shaded regions. The action plots illustrate the outputs of each term on the pyramid stairs (left) and slippery surface (right). Dotted vertical lines in the left figure indicates the instance of toe contact.



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MPC

3 MPC formulations

Humanoid MPC

Optimization on $z = [q^T, \dot{q}^T, F^T]^T$

$$\begin{aligned}
 & \min_{\mathbf{q}[\cdot], \dot{\mathbf{q}}[\cdot], \mathbf{F}[\cdot]} \sum_{i=0}^T (\mathbf{z}_i - \mathbf{z}_{i,des})^T Q_z (\mathbf{z}_i - \mathbf{z}_{i,des}) \Delta t_i \quad (5) \\
 & \text{s.t.} \quad \forall i = 0, \dots, T-1 \\
 & \text{(Initial state)} \quad \mathbf{q}_0 = \bar{\mathbf{q}}_0, \\
 & \quad \quad \quad \dot{\mathbf{q}}_0 = \bar{\dot{\mathbf{q}}}_0, \\
 & \text{(Integration)} \quad \dot{\mathbf{q}}_{i+1} = \dot{\mathbf{q}}_i + \Delta t_i \ddot{\mathbf{q}}_i, \\
 & \quad \quad \quad \mathbf{q}_{i+1} = \mathbf{q}_i + \Delta t_i \dot{\mathbf{q}}_{i+1}, \\
 & \text{(Dynamics)} \quad \mathbf{M}_b(\mathbf{q}_i) \ddot{\mathbf{q}}_i + \mathbf{h}_b(\mathbf{q}_i, \dot{\mathbf{q}}_i) = \mathbf{J}_b(\mathbf{q}_i)^T \mathbf{F}_i, \\
 & \quad \quad \quad \ddot{\mathbf{q}}_i = (\dot{\mathbf{q}}_{i+1} - \dot{\mathbf{q}}_i) / \Delta t_i, \\
 & \text{(Contact)} \quad \|\mathbf{F}_{x,y}\| \leq \mu \mathbf{F}_z \cdot \phi_i, \\
 & \quad \quad \quad \mathbf{J}(\mathbf{q}_i) \dot{\mathbf{q}}_i \cdot \phi_i = 0, \\
 & \text{(Swing)} \quad \mathbf{r}(\mathbf{q}_i)_z = h_{i,des}, \\
 & \text{(Joint limits)} \quad \mathbf{q}_{i,joint} \in \mathcal{Q}, \\
 & \quad \quad \quad \dot{\mathbf{q}}_{i,joint} \in \dot{\mathcal{Q}},
 \end{aligned}$$

- **Input:** $q, \dot{q}$, contact phases ϕ , command $c = [h, v_x, v_y, \omega_z]$.

- **Output:** $\tau_{mpc}, q_{mpc}, \dot{q}_{mpc}$, trajectory z .

TITA MPC

Input: $p_{com}, \dot{p}_{com}, c, v_{cz}, \theta, v, \omega$ Output: a_x, a_z , angular acceleration of the virtual base contact forces

$$\begin{aligned}
 & \min_{\mathbf{u}_0, \dots, \mathbf{u}_{N-1}} \sum_{k=0}^{N-1} \ell(\mathbf{x}_k, \mathbf{u}_k) + \ell_N(\mathbf{x}_N) \\
 & \text{s.t.} \quad \mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) \\
 & \quad \quad \mathbf{x}_0 = \mathbf{x}_{current} \\
 & \quad \quad h(\mathbf{x}_k, \mathbf{u}_k) = 0, \quad g(\mathbf{x}_k, \mathbf{u}_k) \leq 0, \quad k = 0, \dots, N-1 \\
 & \quad \quad h_{ter}^N(\mathbf{x}) = 0 \\
 & \quad \quad \begin{cases} \sum_{i \in I, r} (\mathbf{p}_{c,i} - \mathbf{p}_{com}) \times \mathbf{F}_i = 0 \\ F_{zi} \geq 0 \\ v_{cz} = 0 \\ c_x^N = x_{com}^N \\ c_y^N = y_{com}^N \end{cases}
 \end{aligned}$$

- **Input:** CoM and projected CoM states, orientation, linear and angular velocities.
- **Output:** linear, angular and vertical accelerations;



Tita WBC and trajectory generation

3 MPC formulations

The Whole-Body Controller is formulated as a quadratic program. Given generalized velocity $\mathbf{v} = [\mathbf{v}_b^\top, \boldsymbol{\omega}_b^\top, \dot{\mathbf{q}}_j^\top]^\top$, it includes:

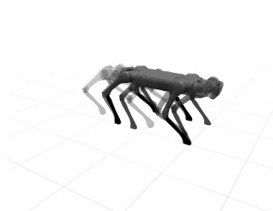
- regularization of velocity, contact forces and torque rate;
- Cartesian-space and joint-space tracking;
- angular-momentum regularization;
- full-body dynamics constraints;
- wheel-ground contact constraints;
- contact-wrench constraints: unilateral contact, friction limits and no-tilting surface conditions.

Trajectory generator: the trajectory is **generated online** at each call of the MPC using virtual unicycle model. **Given a desired velocity**, it integrate a **default acceleration** to converge toward desired command and position.



MPX implementation by IIT

3 MPC formulations



- **MPX** is a library written in **JAX** for GPU array computation. It provides **GPU parallelism**, a fully differentiable solver, and integration into RL environments.
- A **multiple-shooting SQP formulation** solves the **KKT system in parallel**, maintaining exactness and fast convergence. MPX provides **primal-dual** and **FDDP solvers**.
- **JAX** is a Python library for accelerator-oriented array computation, designed for **large-scale machine learning**.
- **Brax** is a framework implementing **RL algorithms in JAX** to train agents.
- MPX also provides **MPC-SRBD** and **MPC-WB** controllers with the **MuJoCo simulator**.



Training information

3 MPC formulations

- **TITA:** MPC at 100 Hz, horizon $N = 50$, $\Delta t_{\text{MPC}} = 0.01$ s; FDDP solver. The inverse-dynamics/WBC layer runs at 100 Hz.
- **Lite3:** MPC at 50 Hz, horizon $N = 25$, $\Delta t_{\text{MPC}} = 0.02$ s; primal-dual solver. WBC runs at 500 Hz.
- Both robots train exclusively on flat terrain and are evaluated on flat, rough terrain with obstacles and Perlin-noise terrains.
- TITA noisy evaluations add zero-mean white noise to joint positions (0.01 rad) and velocities (0.1 rad/s), together with randomized initial conditions.
- The noise affects the measurements provided to both the neural network and MPC, together with randomized initial conditions.



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Tita: residual training setup and evaluation

4 Tita results

- MLP (512, 256, 128), ELU hidden activations; bounded output $a \in [-1, 1]^8$.
- **Outputs:** leg-joint setpoints relative to the default configuration and desired wheel velocities.
- **Observations:** linear/angular velocity, projected gravity, joint-position offsets, joint velocity, CoM-height error, unicycle command, previous action, MPC output and normalized feedforward torque.
- **Given commands during training:**
 - Noise-free training:
 - 70% probability: $v_x \in [0, 2.5]$, $\omega_z \in [0, 0.6]$.
 - 30% probability: $v_x \in [2.5, 3.0]$, $\omega_z \in [0.6, 0.8]$.
 - Noisy training:
 - 80% probability: $v_x \in [0, 2.0]$, $\omega_z \in [0, 0.6]$.
 - 20% probability: $v_x \in [2.0, 2.5]$, $\omega_z \in [0.6, 0.8]$.

$$\tau = \tau_{\text{MPC}} + \tau_{\text{RL}},$$

$$\tau_{\text{RL,leg}} = K_p(q_{\text{default}} + s_{\text{leg}}a_{\text{leg}} - q_{\text{leg}}),$$

$$\tau_{\text{RL,wheel}} = K_d(s_{\text{wheel}}a_{\text{wheel}} + \dot{q}_{\text{wheel}}).$$

$s_{\text{leg}} = 0.5$ rad; $s_{\text{wheel}} = 25$ rad/s. For $r_w = 0.0925$ m, this corresponds to 2.31 m/s maximum wheel speed correction.
Low-level control: 500 Hz; MPC, WBC and policy: 100 Hz.



Tita: PPO training parameters

4 Tita results

- **Training steps:** 30 million residual; 80 million end-to-end
- **Parallel environments:** 1024
- **Episode length:** 1000 control steps
- **Simulation timestep:** 0.002 s
- **Control timestep:** 0.01 s
- **Policy and value MLPs:** (512, 256, 128)
- **Policy distribution:** Tanh-normal
- **Learning rate:** 3×10^{-4}
- **Discount factor:** 0.99
- **Rollout length:** 20 steps per environment between updates
- **Batch size:** 256



Tita: residual reward setup

4 Tita results

- **Linear velocity (10):** $\exp\left(-\|(c_{xy} - v_{xy})/(1 + |c_{xy}|)\|_2^2 / \sigma\right)$
- **Angular velocity (5):** $\exp(-(c_\omega - \omega_z)^2 / \sigma)$
- **Orientation (-1):** $\exp(-\|g_{xy}\|_2^2 / \sigma)$
- **Height (-1):** $\exp(-(c_z - p_z)^2 / \sigma)$
- **First action rate (-0.001):** $\|(a_t - a_{t-1}) / \Delta t\|_2^2$
- **Second action rate (-0.0001):** $\|(a_t - 2a_{t-1} + a_{t-2}) / \Delta t\|_2^2$
- **Joint limits (-1):** $\sum_j [\max(q_j^{\min} - q_j, 0) + \max(q_j - q_j^{\max}, 0)]$
- **Termination (-100):** $\mathbf{1}_{\{\text{fall or base-ground contact}\}}$

c : command; g : projected gravity.



Tita: end-to-end reward setup

4 Tita results

- **Linear velocity (1):** $\exp(-(v_x^{\text{cmd}} - v_x)^2 / 0.0625)$
- **Angular velocity (0.5):** $\exp(-(\omega_z^{\text{cmd}} - \omega_z)^2 / 0.0625)$
- **Orientation (-1):** $\|g_{xy}\|_2^2$
- **Base height (-1):** $1 - \exp(-((h_{\text{com}} - h_{\text{cmd}}) / 0.05)^2)$
- **Posture (-1):** $\exp(-\|c\|_2^2 / 0.25) \sum_j w_j (q_j - q_{j,\text{default}})^2$
- **Joint torques (-0.0001):** $\sum_j \tau_j^2$
- **First action rate (-0.01):** $\|a_t - a_{t-1}\|_2^2$
- **Joint limits (-1):** $\sum_j [\max(q_j^{\text{min}} - q_j, 0) + \max(q_j - q_j^{\text{max}}, 0)]$
- **Termination (-100):** $\mathbf{1}_{\{\text{fall or base-ground contact}\}}$



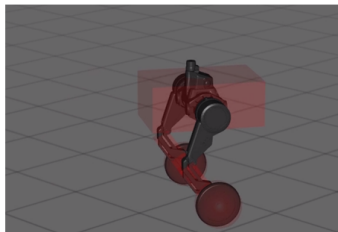
Tita results – terrains

4 Tita results

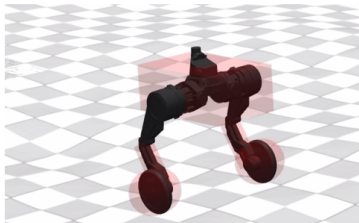
Rough Terrain - 1.5/2.0/2.5 cm



Flat Terrain



Perlin Terrain

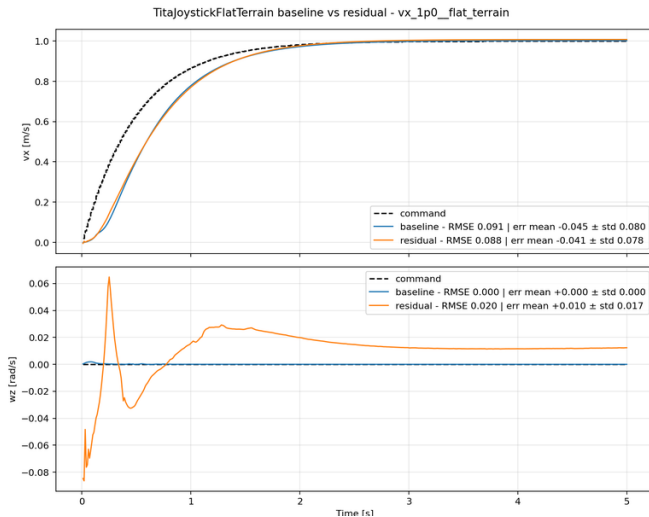




Tita results - flat terrain

4 Tita results

$$v_x = 1.0 \text{ m/s}, \omega_z = 0$$

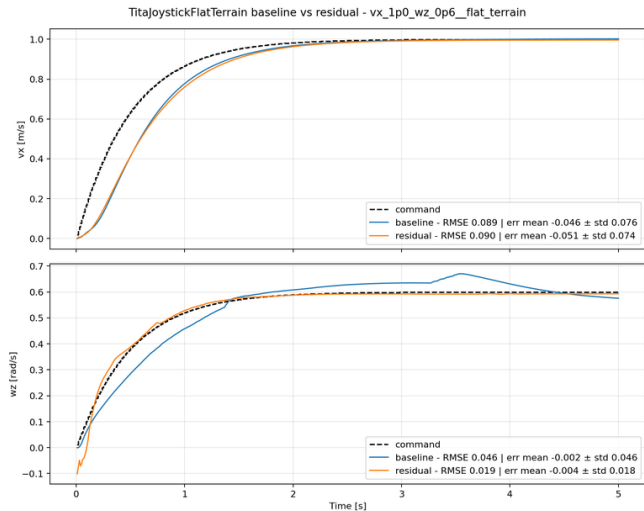




Tita results - flat terrain

4 Tita results

$$v_x = 1.0 \text{ m/s}, \omega_z = 0.6 \text{ rad/s}$$

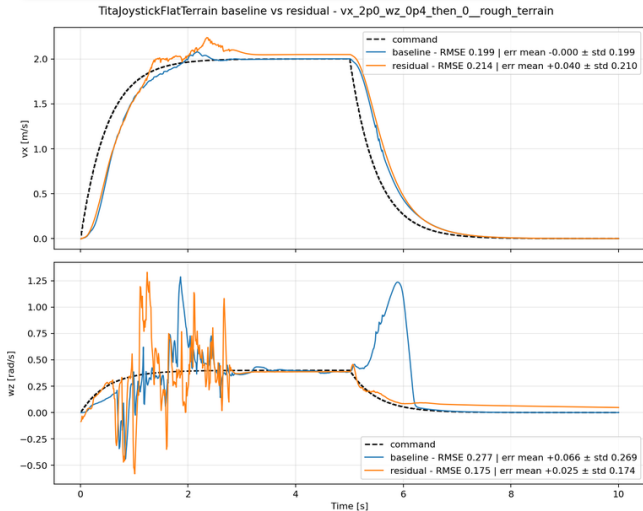




Tita results – rough terrain

4 Tita results

$$(v_x, \omega_z) = (2.0, 0.5) \longrightarrow (0, 0)$$

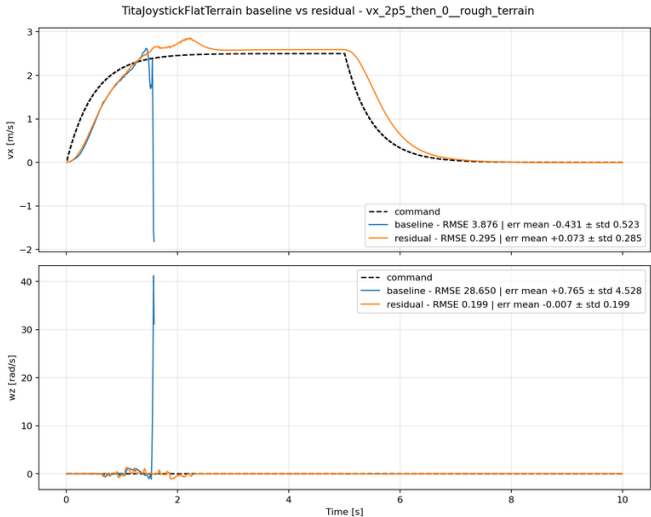




Tita results – rough terrain

4 Tita results

$$v_x = 2.5 \text{ m/s}, \omega_z = 0$$



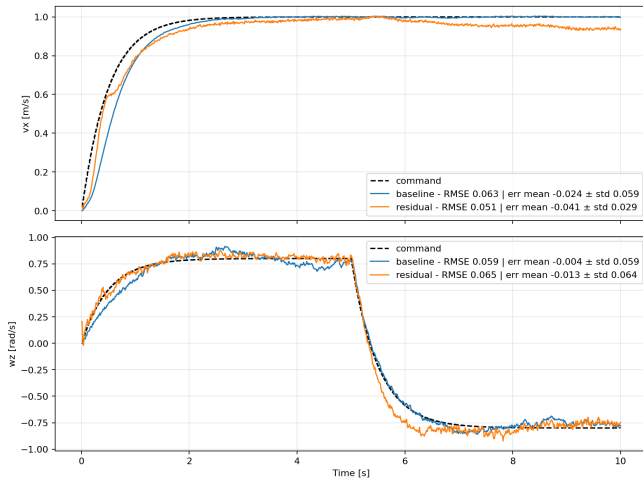


Tita results – flat terrain

4 Tita results

Noisy measurements

TitaJoystickFlatTerrain - flat terrain - velocity tracking
 $vx=1.0$ m/s, $wz=0.8$ rad/s to $vx=1.0$ m/s, $wz=-0.8$ rad/s



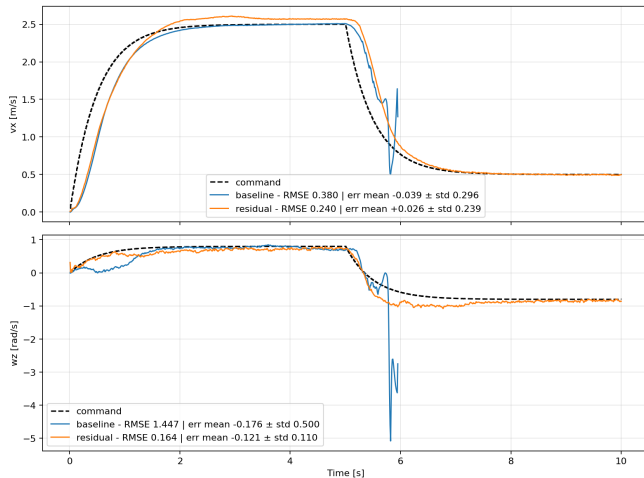


Tita results – flat terrain

4 Tita results

Noisy measurements

TitajoystickFlatTerrain - flat terrain - velocity tracking
 $v_x=2.5$ m/s, $w_z=0.8$ rad/s to $v_x=0.5$ m/s, $w_z=-0.8$ rad/s



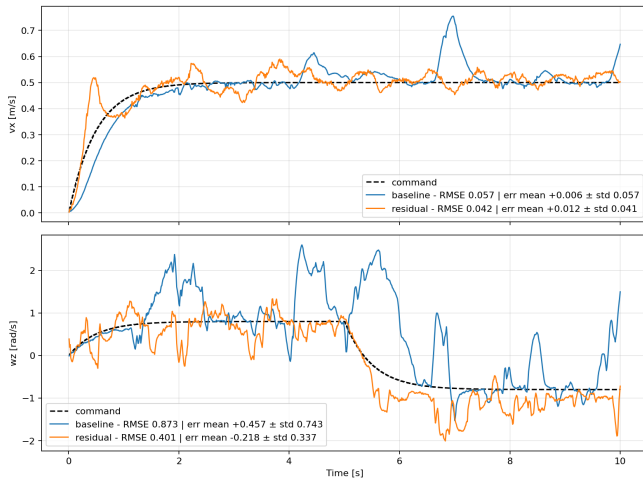


Tita results – rough terrain

4 Tita results

Noisy measurements

TitaJoystickFlatTerrain - rough terrain - velocity tracking
 $v_x=0.5$ m/s, $w_z=0.8$ rad/s to $v_x=0.5$ m/s, $w_z=-0.8$ rad/s



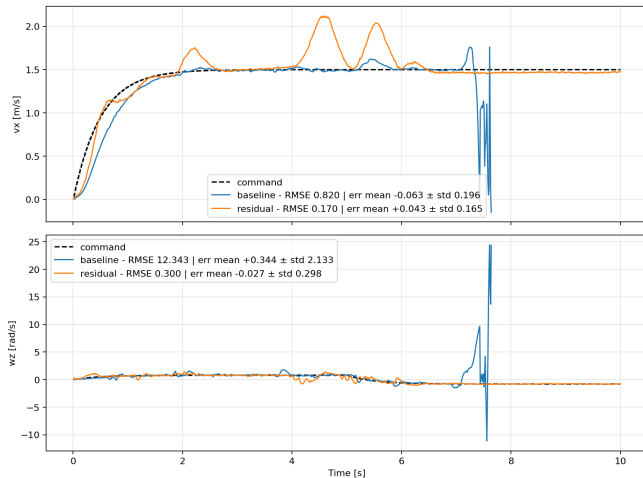


Tita results – perlin terrain

4 Tita results

Noisy measurements

TitaJoystickFlatTerrain - rough terrain - velocity tracking
 $v_x=1.5$ m/s, $w_z=0.8$ rad/s to $v_x=1.5$ m/s, $w_z=-0.8$ rad/s





Tita results – conclusions

4 Tita results

Noise-free evaluation: completed tasks

Task (v_x, ω_z)	Baseline		Residual	
	Flat	Rough	Flat	Rough
(1.0, 0.0)	✓		✓	✓
(1.5, 0.0)	✓	✓	✓	✓
(2.0, 0.0)	✓	✓	✓	✓
(0.0, 0.6)	✓	✓	✓	✓
(0.0, 0.8)	✓	✓	✓	✓
(1.0, 0.6)	✓	✓	✓	✓
(1.5, 0.6)	✓	✓	✓	✓
(2.0, 0.0) → (0.0, 0.0)	✓		✓	✓
(2.5, 0.0) → (0.0, 0.0)	✓	✓	✓	✓
(0.5, 0.8) → (0.5, -0.8)	✓	✓	✓	✓
(1.0, 0.8) → (1.0, -0.8)	✓		✓	✓
(1.5, 0.8) → (1.5, -0.8)	✓		✓	✓
(2.0, 0.4) → (0.0, 0.0)		✓	✓	✓
(2.5, 0.0) → (1.0, 0.8)	✓		✓	✓
(2.0, 0.8) → (1.0, -0.8)	✓		✓	✓
(2.5, 0.8) → (0.5, -0.8)			✓	
(1.0, 0.8) → (-1.0, -0.8)	✓		✓	✓
(2.0, 0.8) → (-1.0, -0.8)			✓	
Total — Flat	15/18		18/18	
Total — Rough		9/18		16/18



Tita results – conclusions

4 Tita results

Additional evaluation: completed tasks

Task (v_x, ω_z)	Baseline	Residual
(0.2, 0.0)	✓	✓
(0.5, 0.0)		
(0.7, 0.0)		✓
(1.0, 0.0)		
(0.0, 0.5)	✓	✓
(0.5, 0.3)		
(0.7, 0.3)		
(1.0, 0.3)	✓	✓
(1.0, 0.0) $\rightarrow$ (0.0, 0.0)		
(0.5, 0.3) $\rightarrow$ (0.5, -0.3)		
(0.7, 0.3) $\rightarrow$ (0.7, -0.3)		
(1.0, 0.3) $\rightarrow$ (0.5, -0.3)		
Total	3/12	4/12



Tita results – conclusions

4 Tita results

- In both noise-free and noisy conditions, the neural network increases the maximum achievable velocity and improves locomotion robustness across different terrains.
- However, the neural network can introduce sometimes disturbances into the nominal locomotion, resulting in less natural motion, for example, a wider leg stance and orientation oscillations.
- Those problems can be fixed with some action-filter or with better reward shaping



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Lite3: residual training setup and evaluation

5 Lite3 results

- MLP (256, 256, 256) with ELU activations and bounded actions $a \in [-1, 1]$ ¹².
- **Outputs:** joint setpoints relative to the default configuration.
- **Observations:** linear/angular velocity, projected gravity, joint-position offsets, joint velocity, unicycle command and previous action.
- **Training commands:** $v_x \in [0, 1.5]$, $v_y \in [0, 0.6]$ m/s and $\omega_z \in [0, 1.2]$ rad/s.

$$\tau = \tau_{\text{MPC}} + K_p(q_{\text{default}} + sa - q) - K_d\dot{q}, \quad s = 0.5.$$

- **SRBD MPC inputs:** base pose and velocity, velocity/height commands, foot positions and measured contacts.
- **Outputs:** predicted base trajectory and optimal ground-reaction forces.
- **Tasks:** height tracking, roll/pitch stabilization, velocity tracking, contact-force optimization and friction-cone enforcement.
- WBC map ground-reaction-force into joints.



Lite3: residual reward setup

5 Lite3 results

- **Linear velocity (1):** $\exp(-\|v_{xy}^{\text{cmd}} - v_{xy}\|_2^2 / 0.25)$
- **Angular velocity (0.5):** $\exp(-(\omega_z^{\text{cmd}} - \omega_z)^2 / 0.25)$
- **Orientation (-5):** $\|(z_{\text{torso}})_{xy}\|_2^2$
- **Vertical velocity (-0.5):** v_z^2
- **Roll/pitch velocity (-0.05):** $\omega_x^2 + \omega_y^2$
- **Default pose (0.5):** $\exp(-\sum_j w_j (q_j - q_{j,\text{default}})^2)$
- **Joint limits (-1):** $\sum_j [\max(q_j^{\text{min}} - q_j, 0) + \max(q_j - q_j^{\text{max}}, 0)]$
- **Termination (-10):** $\mathbf{1}_{\{\text{fall or non-finite state}\}}$
- **Residual torque (-0.02):** $\frac{1}{12} \sum_j \tau_{\text{res},j}^2$
- **Joint saturation (-0.5):** $\text{mean}_j(\mathbf{1}_{\{j \text{ saturated}\}})$
- **First action rate (-0.01):** $\|a_t - a_{t-1}\|_2^2$
- **Second action rate (-0.001):** $\|a_t - 2a_{t-1} + a_{t-2}\|_2^2$
- **Joint torques (-0.0002):** $\sqrt{\sum_j \tau_j^2} + \sum_j |\tau_j|$
- **Energy (-0.001):** $\sum_j |\dot{q}_j| |\tau_j|$



Lite3 results – terrains

5 Lite3 results

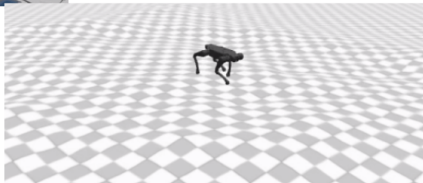
Rough Terrain



Flat Terrain



Perlin Terrain

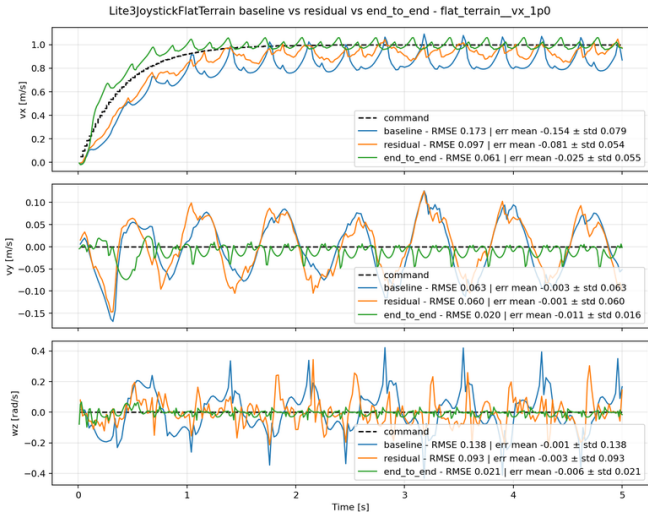




Lite3 results - flat terrain

5 Lite3 results

$$v_x = 1.0 \text{ m/s}, \quad v_y = 0.0 \text{ m/s}, \quad \omega_z = 0.0 \text{ rad/s}$$



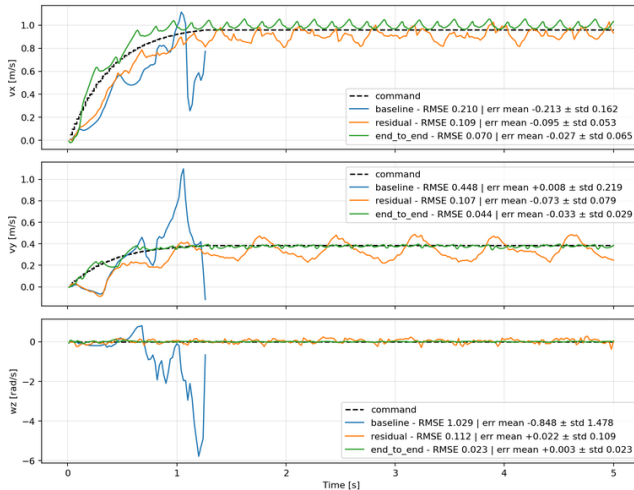


Lite3 results - flat terrain

5 Lite3 results

$$v_x = 1.0 \text{ m/s}, \quad v_y = 0.0 \text{ m/s}, \quad \omega_z = 0.6 \text{ rad/s}$$

Lite3joystickFlatTerrain baseline vs residual vs end_to_end - flat_terrain_vx_1p0_vy_0p4



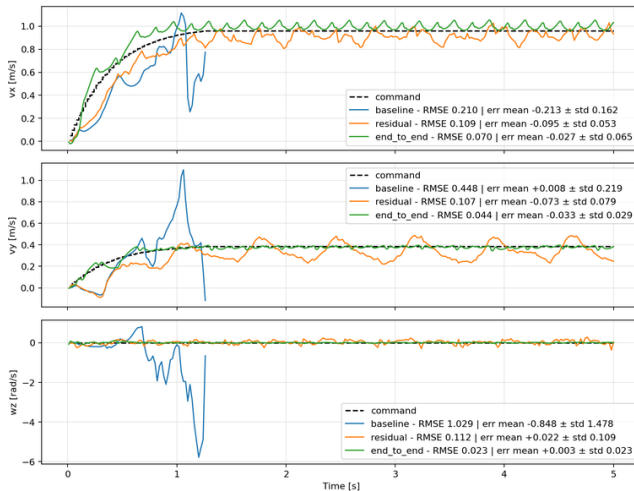


Lite3 results - flat terrain

5 Lite3 results

$$v_x = 1.0 \text{ m/s}, \quad v_y = 0.4 \text{ m/s}, \quad \omega_z = 0.0 \text{ rad/s}$$

Lite3joystickFlatTerrain baseline vs residual vs end_to_end - flat_terrain_vx_1p0_vy_0p4



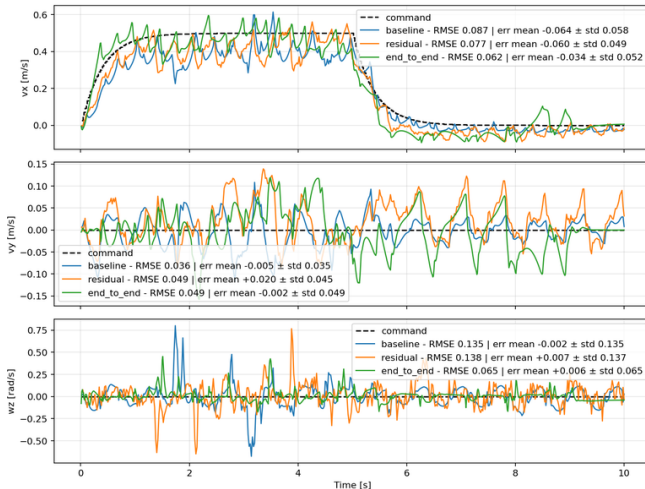


Lite3 results - rough terrain

5 Lite3 results

$$v_x : 0.5 \rightarrow 0.0 \text{ m/s}, \quad v_y = 0.0 \text{ m/s}, \quad \omega_z = 0.0 \text{ rad/s}$$

Lite3JoystickFlatTerrain baseline vs residual vs end_to_end - rough_terrain_vx_0p5_then_0





Lite3 results - perlin terrain

5 Lite3 results

$$v_x = 1.0 \text{ m/s}, \quad v_y = 0.0 \text{ m/s}, \quad \omega_z = 0.6 \text{ rad/s}$$

Lite3JoystickFlatTerrain baseline vs residual vs end_to_end - perlin_terrain_vx_1p0_wz_0p6

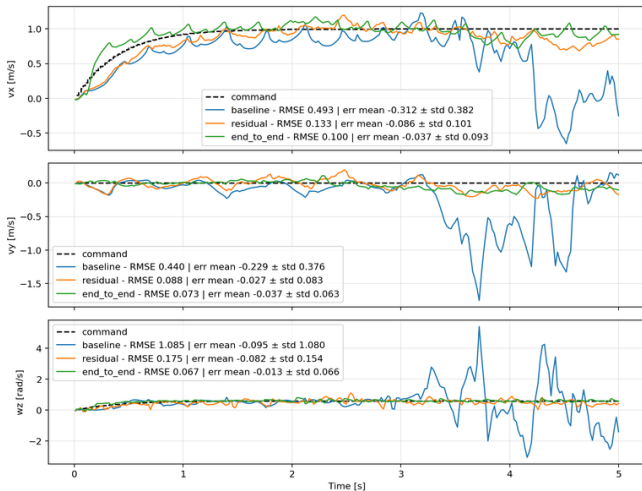




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Conclusions and contributions

6 Conclusions and contributions

- This work **extends the MPX library with a simulation environment for wheeled-legged robots**, focusing on the bipedal TITA platform.
- A **constraint-aware Whole-Body Controller** is formulated as a **Quadratic Program**, replacing algebraic inverse dynamics.
- The QP explicitly accounts for **full-body dynamics**, wheel-ground contact constraints, friction limits, contact-wrench feasibility, and actuator limits.
- **Residual learning is integrated with an MPC-WBC pipeline**, allowing the neural network to **correct the joint commands** generated by the nominal controller.
- This work investigates a **less explored residual-learning architecture combining MPC with a separate QP-based WBC layer**. Existing approaches generally include dynamics directly in MPC or rely on algebraic inverse dynamics.
- The approach is evaluated on **two robotic platforms**, a quadruped and a bipedal wheeled-legged robot, across **unseen terrains**, with additional **noisy tests for TITA**.
- The residual policy can **reduce tracking errors, increase the maximum sustainable velocity, and improve robustness** on rough and Perlin-noise-generated terrains.
- Future work: **improved reward design, action filtering, and real-robot validation**.